A PROJECTIVE SCHEME AND A FINITE TYPE QUASI-COHERENT MODULE ON P^1 WITH NON-FINITELY GENERATED H^0

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds no mathematics of its own; the bidegree table at 5299-5313 is the pivot of the kernel computation.

Editorial note. The label lemma-nonfinite-h0 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 5280-5285: R any non-Noetherian ring, I an ideal that is not finitely generated, S = R[x, y]/xyI graded with x, y of degree 1; X = Proj(S) is a projective scheme over R; the claim is that H^0(X, O_X) is not a finitely generated R-module
- Context span 5287-5325: decomposing S_x, S_y, S_{xy} by bidegree gives summands R'_{a,b} (R for b = 0, R/I for b != 0), R"_{a,b} (R for a = 0, R/I for a != 0), and R/I in every bidegree; H^0(X, O_X) is the kernel of (S_x)_0 x (S_y)_0 -> (S_{xy})_0, (F, G) -> F - G, which by the explicit descriptions is the submodule M of R^2 of pairs with F - G in I; the surjection M -> I given by (F, G) -> F - G shows M is not finitely generated
- Context span 5327-5332: the surjection R[x, y] -> S gives a closed immersion i: X -> P^1_R; F = i_*O_X is a finite type quasi-coherent module on P^1_R whose H^0 equals the non-finitely generated module M

Let R be any non-Noetherian ring and I an ideal of R that is not finitely generated. Let $S = R[x, y]/xyI$ viewed as a graded ring with x and y having degree 1. Then $X = \text{Proj}(S)$ is a projective scheme over R. We claim that $H^0(X, \mathcal{O}_X)$ is not a finitely generated R-module.

Decomposing the rings S_x, S_y, and S_{xy} by bidegree, we can write

$$S_x = \bigoplus_{\substack{(a,b)\in\mathbf{Z}^2 \\ b\geq 0}} R'_{a,b}, \qquad S_y = \bigoplus_{\substack{(a,b)\in\mathbf{Z}^2 \\ a\geq 0}} R''_{a,b}, \qquad S_{xy} = \bigoplus_{(a,b)\in\mathbf{Z}^2} R/I,$$

where the (a, b)-summands sit in degree $a + b$ and we have

$$R'_{a,b} = \begin{cases} R & b = 0, \\ R/I & b \neq 0, \end{cases} \qquad R''_{a,b} = \begin{cases} R & a = 0, \\ R/I & a \neq 0. \end{cases}$$

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The R-module $H^0(X, \mathcal{O}_X)$ can be computed as the kernel of the map

$$(S_x)_0 \times (S_y)_0 \to (S_{xy})_0$$

defined by $(F, G) \mapsto F - G$. Here the notation A_0 refers to the degree 0 component of a $\mathbf{Z}$-graded ring A. By our explicit description of S_x, S_y, S_{xy} above, this kernel is isomorphic to the submodule $M \subset R^2$ consisting of those pairs (F, G) with $F - G \in I$. Since the map $M \to I$ of R-modules given by $(F, G) \mapsto F - G$ is surjective, M is not a finitely generated R-module.

The surjection $R[x, y] \to S$ corresponds to a closed immersion $i : X \to \mathbf{P}^1_R$. Then $\mathcal{F} = i_*\mathcal{O}_X$ is a finite type quasi-coherent $\mathcal{O}_{\mathbf{P}^1_R}$-module whose H^0 is equal to the non-finitely generated module M we computed above.

Lemma 0.1. *Non-finite H^0.*

(1) *There exists a ring R and a projective scheme X over R such that $H^0(X, \mathcal{O}_X)$ is not a finite R-module.*
(2) *There exists a ring R and a finite type quasi-coherent module $\mathcal{F}$ on $\mathbf{P}^1_R$ such that $H^0(\mathbf{P}^1_R, \mathcal{F})$ is not a finite R-module.*

Proof. See discussion above. □